

AssureX Claim Engine — Project Report

NextWave AI & ML · Aptech Limited · Version 2.0 (September 2026)

1. Problem definition

Manufacturers and service centers receive warranty claims that must be checked against purchase proof, product age, warranty terms, exclusions, repair history and supporting evidence. Done by hand this is slow, inconsistent between staff, and easy to defeat with reused receipts, altered dates or mismatched serial numbers. AssureX automates the checking and produces a recommendation — Likely Valid, Likely Invalid or Manual Review Required — that is explained, audited and sent to a person whenever the evidence is uncertain.

2. Background and business necessity

Warranty cost is a direct expense for manufacturers and a satisfaction driver for customers. Typical failure modes of manual review are: approving claims filed after the warranty and its grace period, missing exclusions (drops, liquid damage), accepting the same invoice twice, and inconsistent decisions for identical cases. An automated first pass that applies the policy the same way every time, flags contradictions and duplicates, and routes only the unclear cases to reviewers shortens turnaround and makes decisions defensible.

3. Proposed solution

A Python/Flask web application in which:

- customers (or service-center staff for walk-in customers) register products and warranties and file claims with receipts, warranty cards, photos, videos and diagnostic reports;
- documents are validated, fingerprinted (SHA-256) and read by OCR; the user confirms extracted values;
- the claim is converted into a structured feature record and scored by a Python classification model;
- the same record is drawn as a Claim Summary Card image and scored independently by a Google Teachable Machine model;
- a rule engine checks the category's warranty policy; detectors look for contradictions and duplicates;
- a configurable decision table combines rules, both predictions and their consistency status into the final recommendation; uncertain claims enter a manual-review queue;
- reviewers decide (with overrides that require a reason), customers track progress through eight stages, and administrators see dashboards, analytics, exports and a full audit trail.

4. Purpose of the document

This report describes the requirements, design, data, models, testing and limitations of AssureX so that evaluators, maintainers and stakeholders share one understanding of what was built and how to verify it.

5. Scope

In scope: registration and authentication for four roles, product/warranty records, receipt upload and OCR, claim intake, evidence management, repair history, validation and pre-processing, the common dataset, both models, comparison, warranty rules, contradiction/duplicate/missing-document detection, claim summary and explanation, manual review, tracking, notifications and alerts, dashboards, search, analytics, PDF reports, CSV export, audit trail, model versioning, error handling and monitoring. Out of scope (SRS §1.4): live manufacturer databases, payment systems, enterprise warranty platforms.

6. Assumptions

- Claims arrive through the web interface; a claim concerns one registered product with one warranty.
- Service-center staff belong to one service center; products may name a preferred authorised center.
- The warranty policy for a category is the JSON file in [policies/](#); changing it changes future evaluations only.
- Diagnostic confidence (0–1) comes from a technician; if nobody assessed the fault it is 0.5 (neutral).
- The training data is synthetic, generated from the policy with 4% label noise.

7. Constraints

- Quality depends on the completeness and honesty of the submitted evidence (SRS §1.5).
- The dataset size (1,500 claims) limits how much rare scenarios can be learned (e.g. late reporting).
- The two models are trained independently and do not produce identical confidences; the comparison thresholds absorb normal differences.
- Teachable Machine training happens in the browser and must be repeated by hand when cards change.
- Personal data (names, contact details, receipts) must be protected: role-scoped access, no public file URLs, audit logging, and demo data only in the repository.

8. Functional requirements (SRS §1.6) — where each is implemented

#	Requirement	Implementation
i	Registration & authentication, role-based access	src/api/auth.py , src/security/ , config/rbac.json — customer self-sign-up, other roles invited
ii	Profile, unique User ID	/profile , User.user_id (USR-...)
iii	Product registration, Product ID	/products/new , Product.product_id (PRD-...)
iv	Standard & extended warranties, provider, dates, conditions, exclusions, service center; active/expired/expiring view	ProductWarranty , product page, product list filters
v	Upload receipts/invoices/warranty cards (PDF/JPG/JPEG/PNG)	src/services/documents.py (magic-byte validation)
vi	Extract purchase date, invoice, product, model, serial, retailer, amount, warranty length	src/ocr/document_processor.py
vii	Show extracted values, let users correct them	Wizard step 3, product form auto-fill, claim page Extracted details (audited)
viii	Warranty dates, remaining period, active/expired/expiring/extended	ProductWarranty.status , life-used bars
ix	Expiry alerts with admin-configured window	src/services/alert_service.py , Admin > Policies > alerts
x	Claim registration by users or staff, Claim ID, links	/claims/new , Claim.claim_id (CLM-...)
xi	Claim information (age, purchase, fault date, description, cause, service history, replacement, submission date)	Wizard step 2, Claim model
xii	Photos, damage images, videos, serial photos, diagnostic reports	Eight evidence types incl. MP4 video
xiii	Repair history (dates, center, parts, outcome, cost, authorised or not)	RepairHistory , product page (staff)
xiv	Documents organised per product and claim; view/download/replace/remove by rights	Product and claim evidence panels, document.* permissions

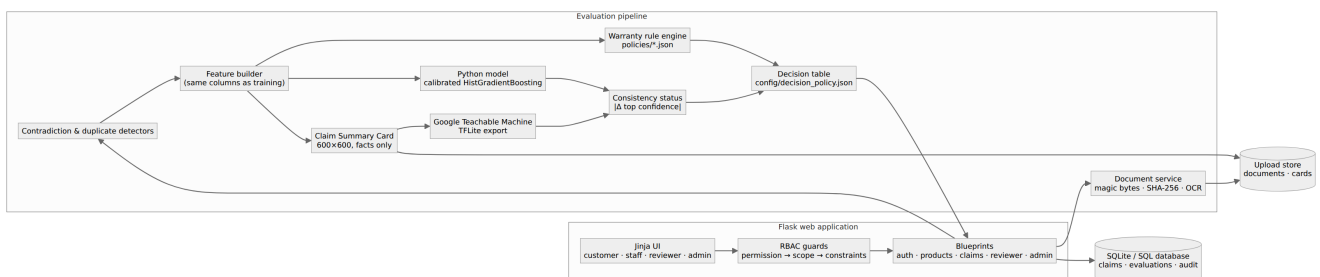
#	Requirement	Implementation
xv	Validation: mandatory fields, dates, numbers, file types, sizes, duplicate IDs	src/rules/validator.py , documents.validate , unique DB constraints
xvi	Pre-processing: missing values, dates, encoding, normalisation, derived fields	src/core/features.py + saved sklearn pipeline
xvii	Common dataset in CSV and card form	data/ , dataset_generator/
xviii	Train & compare algorithms, select best	notebooks/train_python_v2.py
xix	Python confidence for all three classes	Claim page, stored per evaluation
xx	Claim Summary Card without prediction/decision	src/core/card_v2.py
xxi	Teachable Machine classification	src/core/gtm_classifier_v2.py
xxii-xxiv	Class match, [Δ confidence], five consistency statuses with configurable thresholds	src/core/consistency.py , config/decision_policy.json
xxv-xxvi	Warranty rule validation, configurable per category	src/rules/policy_engine.py , policies/*.json , Admin > Policies
xxvii	Serial number vs receipt, warranty card, product image, repair records	src/rules/contradiction_detector.py
xxviii	Contradictions (dates, models, serials)	same
xxix	Missing mandatory documents, telling the user which	Policy mandatory_documents / supporting_documents , checklist, notifications
xxx-xxxi	Duplicate claims and documents (IDs, invoices, serials, descriptions, claimant, hashes, history)	src/rules/duplicate_detector.py
xxxii	Claim summary	src/services/explain.py → summarise (template-based, no external AI)
xxxiii	Claim preparation assistance	Live readiness checklist in the wizard
xxxiv	Final decision from all factors	src/core/pipeline.py , decision table
xxxv	Explanation: supporting/opposing factors, rules passed/failed, contradictions, evidence needed	explain.py → explain, claim page
xxxvi-xxxvii	Manual-review queue, approve/reject/request info, comments, override with reason, original AI result kept	src/api/reviewer.py , ReviewerAction
xxxviii	Eight tracked stages	Claim.transition_status , stage bar, tracking page
xxxix	Notifications (expiry, submission, missing docs, info requests, status, review, approval, rejection)	src/services/notifications.py , bell menu, dashboard
xl	User dashboard	/claims/
xli	Admin dashboard (totals, classes, pending, duplicates, disagreements, average confidences, trend)	/admin/dashboard
xlII	Search & filter by claim, product, category, serial, warranty status, claim status, risk, confidence, reviewer, dates	/claims/search
xlIII	Analytics (outcomes, faults, rejection reasons, categories, expirations, repairs, model performance, manual-review frequency)	/admin/analytics
xliv	Downloadable claim report	/claims/<id>/report.pdf (src/services/report_generator.py)
xlV	CSV/Excel export	/admin/export/<kind> (CSV with BOM, or ?format=xlsx workbook); formula-injection guard in both
xlvi	Secure database storage	SQLAlchemy models, SQLite/any SQL
xlVII	Audit trail	AuditLog , src/services/audit.py , Admin > Audit
xlVIII	Model version tracking; updates don't alter recorded results	Hash-based versions on every ModelEvaluation row; rows never updated
xlIX	Understandable errors	Friendly error pages and flash messages; model/OCR failures degrade gracefully

#	Requirement	Implementation
I	Monitoring & anomaly alerts	<code>alert_service.anomalies()</code> on the admin dashboard; Send due alerts

9. Non-functional requirements

NFR	Result
Performance ≤ 5 s	Full evaluation (rules + Python + card + image model + decision) ~150–300 ms on a laptop; stored per claim (<code>latency_ms</code>) and tested
Scalability (10,000 claims, several centers, concurrent users)	Indexed columns (claim_id, status, reviewer, serial, hashes); scoped SQL queries; stateless workers; any SQL database via <code>DATABASE_URL</code>
Usability and accessibility	Four-step wizard with a live readiness checklist and server-saved drafts, plain-language explanations, responsive layout from 390 px, keyboard shortcuts for reviewers, visible focus, reduced-motion support. axe-core finds no serious or critical WCAG 2 A/AA issue on 21 pages; Lighthouse scores 100 for accessibility, best practices and SEO and 99 for performance on the public pages
Accuracy $\geq 85\%$	Python model 89.3%, Teachable Machine 93.8% on the 225 unseen test claims
Availability $\geq 99\%$	Stateless app behind gunicorn; <code>/healthz</code> for uptime monitors; no runtime dependency on external APIs or CDNs

10. Application architecture



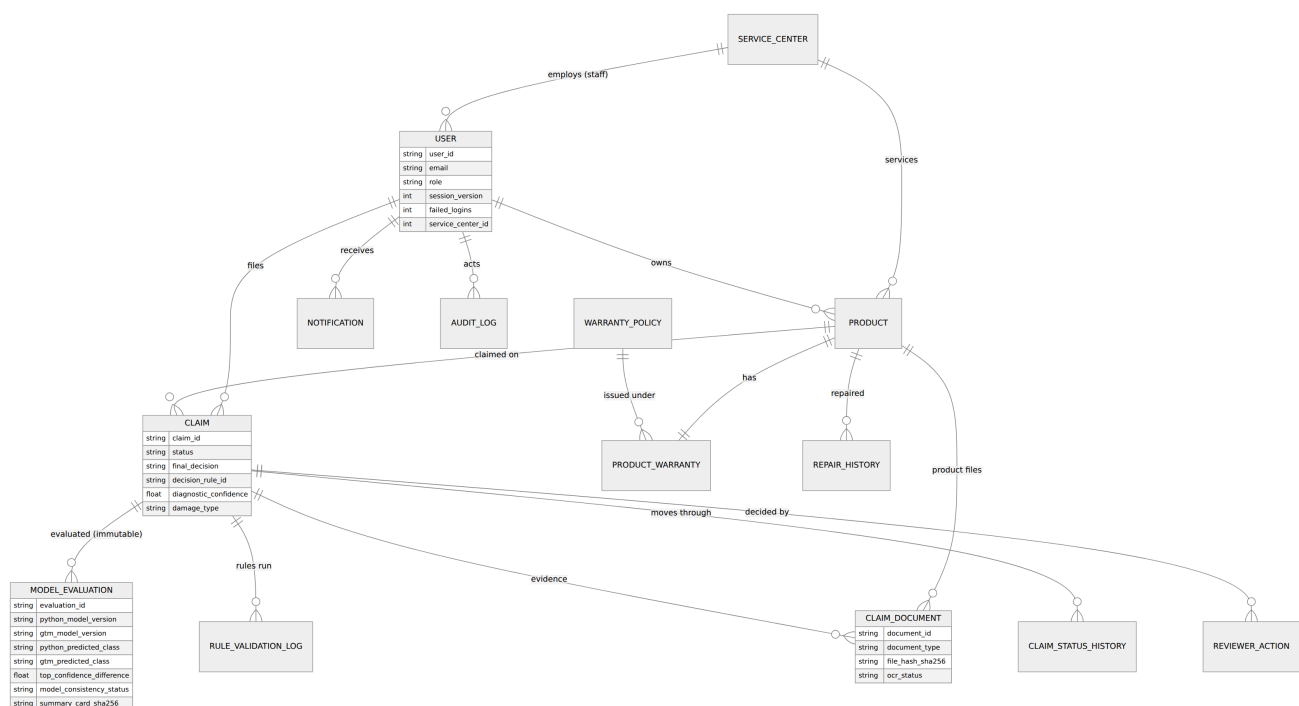
Three layers: the web layer (Jinja templates on one component library, design tokens in `static/css/tokens.css`, self-hosted IBM Plex fonts, a shell script plus small page scripts and the Alpine.js CSP build; the Content-Security-Policy allows only same-origin scripts and one nonce'd bootstrap line, and every JSON endpoint answers with one `{success, data, error}` envelope), the application layer (Flask blueprints guarded by the RBAC engine; services for the claim lifecycle, documents, alerts, analytics, exports and reports) and the evaluation layer (feature builder, rule engine, detectors, both model runtimes, card renderer, consistency and decision table). Persistence is SQLAlchemy over SQLite by default plus a file store for documents and cards.

11. Module descriptions

Module	Responsibility
<code>src/core/vocab.py</code>	Single vocabulary: categories, damage types, faults, classes, document types, date formats
<code>src/core/features.py</code>	Pre-processing and feature engineering for a stored claim
<code>src/core/python_classifier.py</code>	Loads the versioned sklearn pipeline; never retrains itself
<code>src/core/card_v2.py</code>	Deterministic 600×600 card renderer (v3 tile design) with bundled fonts
<code>src/core/gtm_classifier_v2.py</code>	Teachable Machine TFLite runtime, label parsing, version hashing

Module	Responsibility
src/core/consistency.py , decision_table.py	Five statuses; first-match decision table with validation and safe saving
src/core/pipeline.py	Orchestration and immutable persistence of each evaluation
src/core/offline_eval.py	Same pipeline over the val/test splits for evaluation and reports
src/rules/policy_store.py	Loads/validates/saves policy JSON; rule catalogue
src/rules/policy_engine.py	Sixteen rule checks; severity from policy
src/rules/contradiction_detector.py , duplicate_detector.py	Evidence consistency and duplicate indicators
src/rules/validator.py	Form validation and preparation checklist
src/security/rbac.py , guards.py	Permission → scope → constraint engine; request hooks, decorators, scoped queries, login hardening
src/services/*	Claim lifecycle, documents, notifications, audit, alerts, analytics, exports, PDF report, summaries
src/api/*	Public, auth, products, claims, reviewer (queue and workbench), admin, access-control blueprints
src/api/api.py , errors.py	JSON endpoints (stored verdict, live bench, draft autosave, what-if, batch) and the error vocabulary
src/core/explain_models.py	Per-claim explanations: Python feature contributions (replace each input with its training median/mode) and Teachable Machine tile occlusion
src/services/verdict.py	Verdict screen data from the stored payload: agreement-meter geometry, reviewer overrides, customer visibility
src/services/whatif.py , batch.py	Threshold simulator over the labelled test claims and live claims; chunked batch evaluation of uploaded CSVs
src/services/model_card_service.py	Public model card: metrics, calibration (reliability bins, ECE), per-category accuracy, label-noise analysis

12. Database design



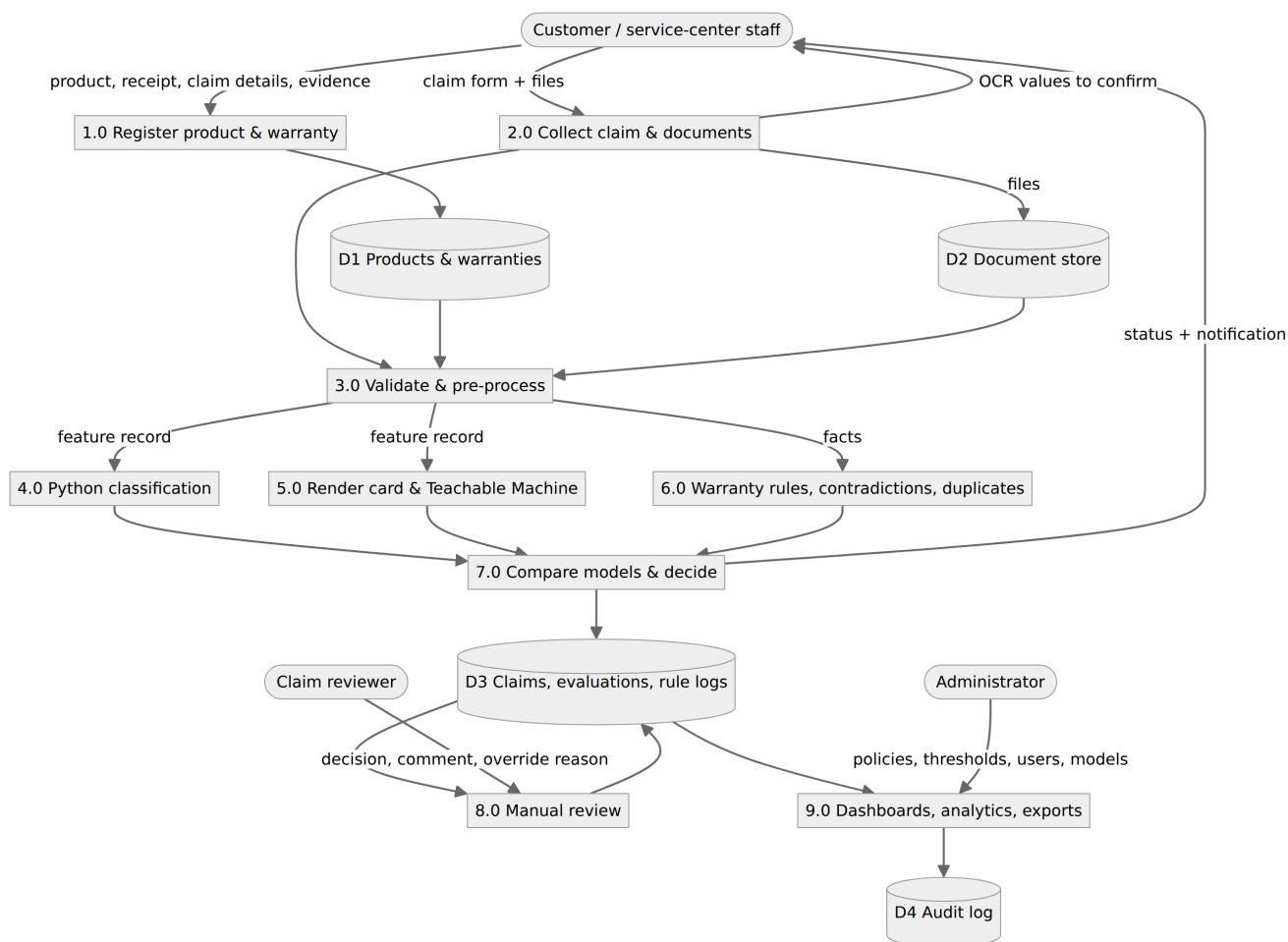
Fifteen tables. Key points: public identifiers (USR-, PRD-, CLM-, DOC-, EVL-) are random and unique; **claims** carries status, recommendation, decision rule and flags; **model_evaluations** stores the exact

feature record, both predictions with all three confidences, model versions, card hash, thresholds and the decision trace — one row per run, never updated; [rule_validation_logs](#) stores every rule with its severity and message plus contradictions, duplicate indicators and missing documents; [claim_status_histories](#) and [audit_logs](#) are append-only; [users](#) hold security state (session version, failed logins, lockout, forced password change, service center).

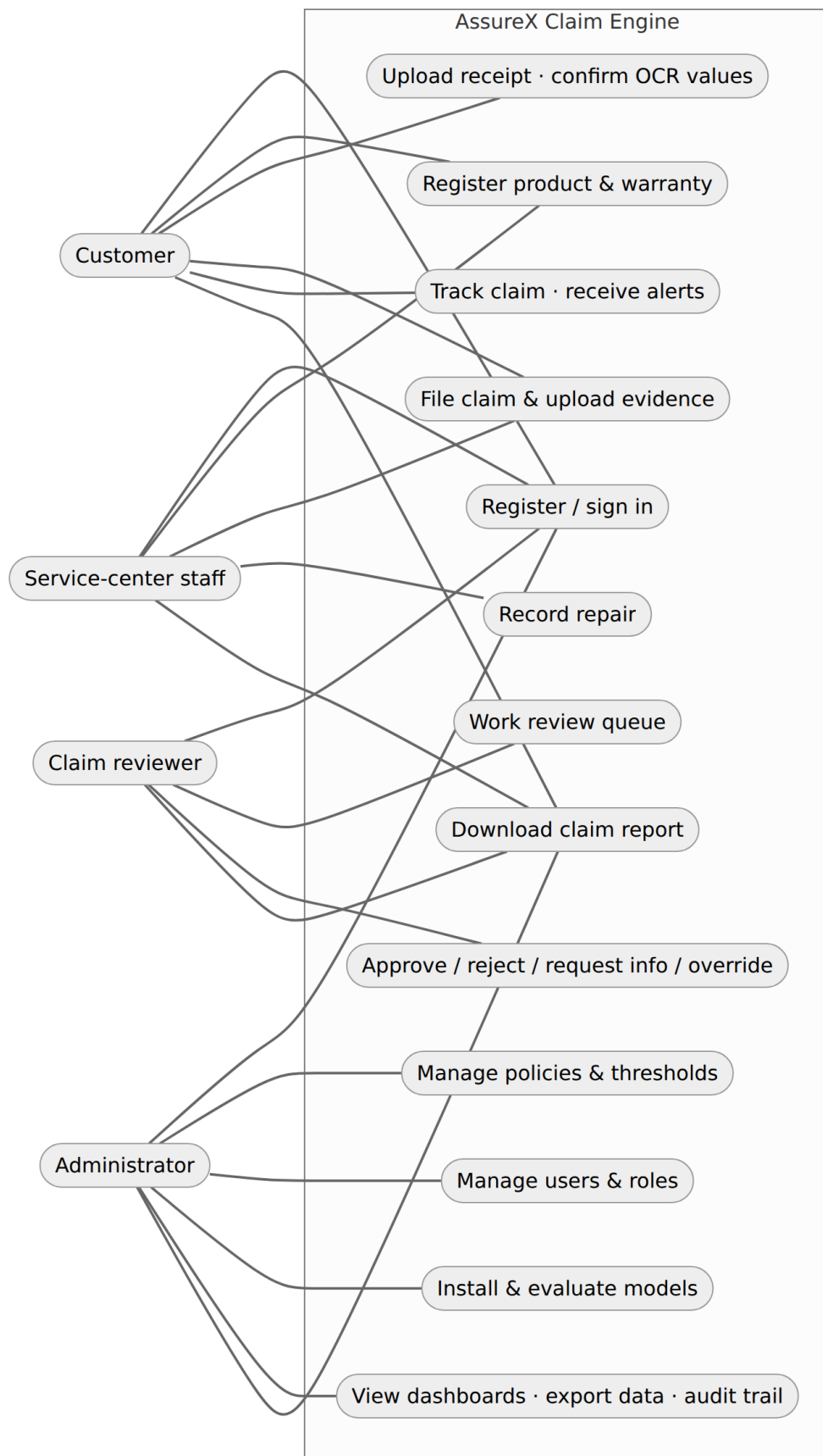
13. Data dictionary

The dataset columns are documented in [data/data_dictionary.json](#) (37 columns: identifiers, descriptive fields, 20 model features, scenario and label). Database fields are documented in [src/models/entities.py](#).

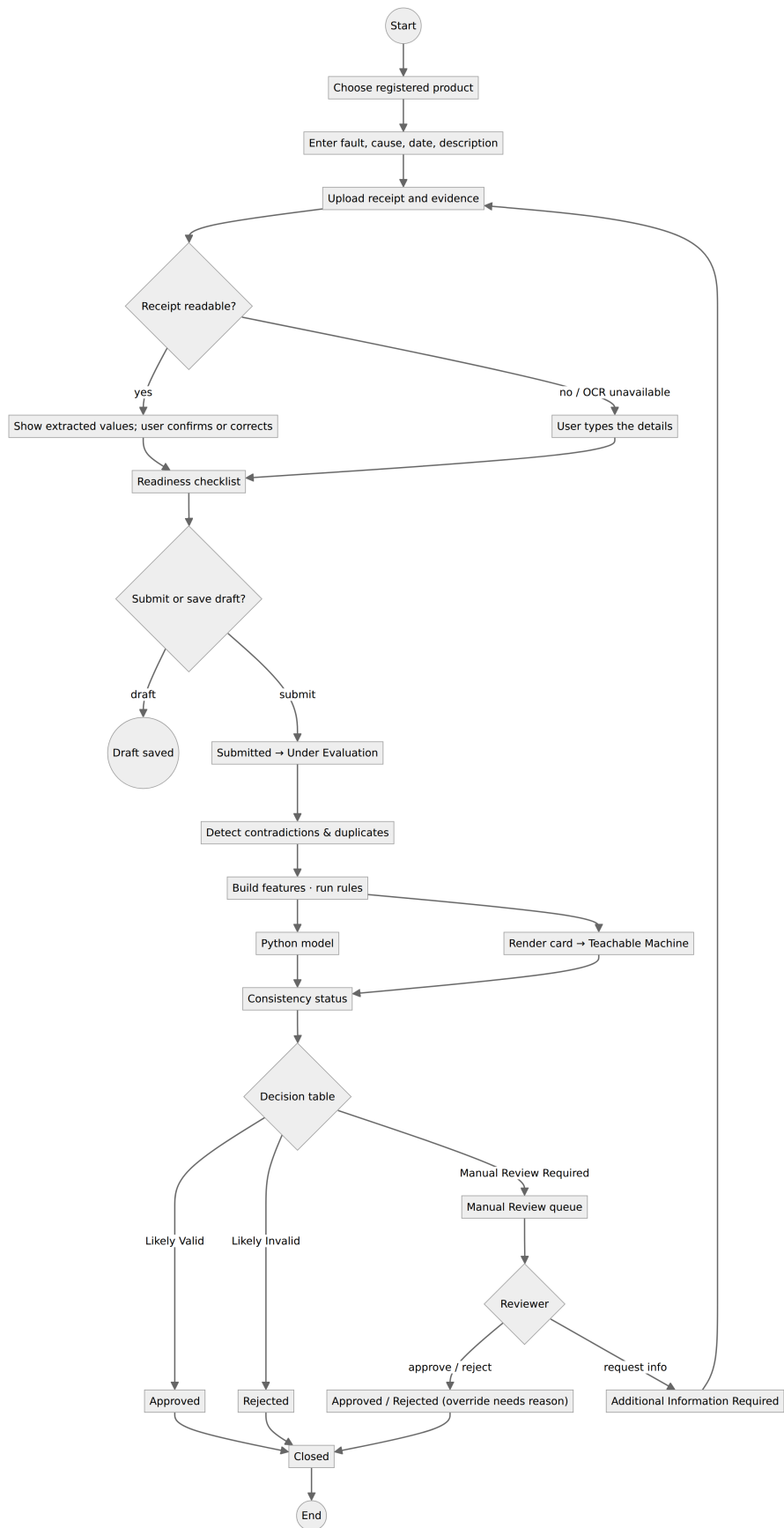
14. Data flow diagram



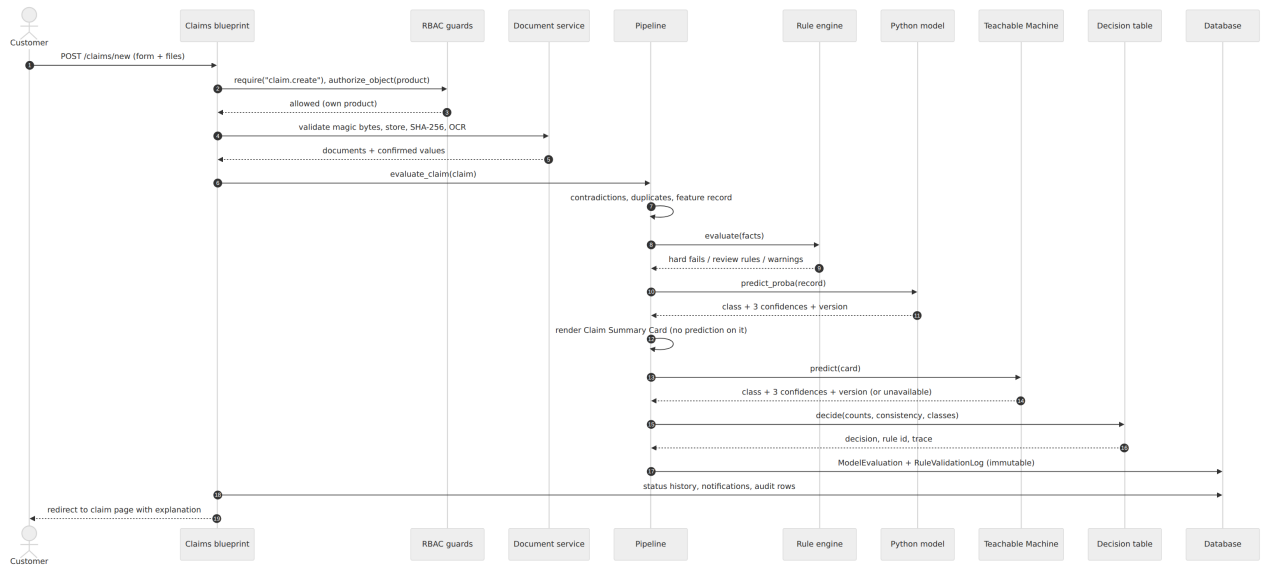
15. Use case diagram



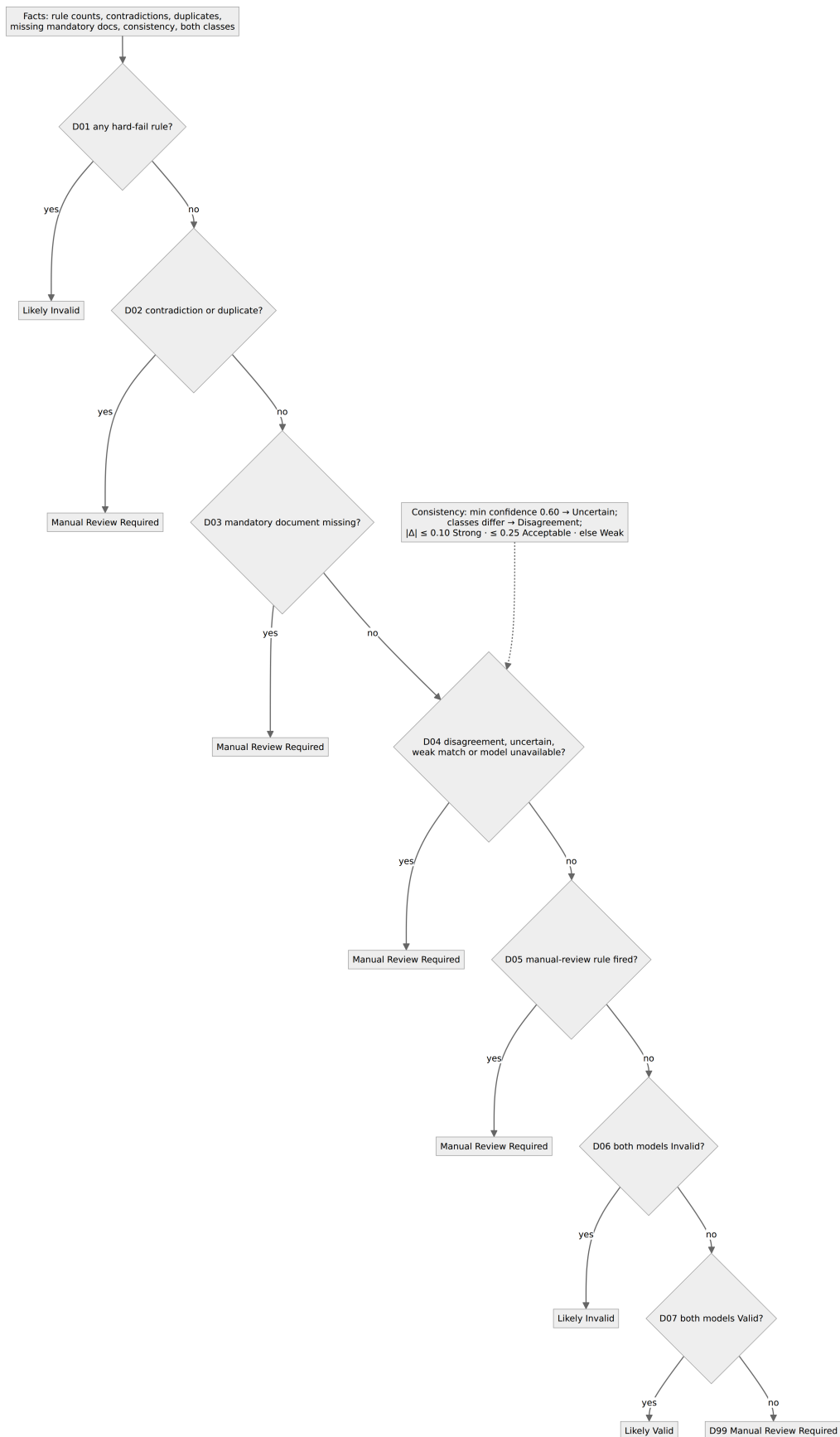
16. Activity diagram



17. Sequence diagram



18. Decision flow diagram



19. Rule-engine design

Each category has a JSON policy with coverage, standard terms, extension length, start conditions, grace period, reporting period, covered faults, excluded damage causes (plus human-readable exclusions), diagnosis threshold for confirming an exclusion, repeat-repair threshold, repair and

replacement conditions, authorised-center requirement, mandatory and supporting documents, and three rule lists: [hard_fail_rules](#), [manual_review_rules](#), [warning_rules](#). The engine has sixteen checks (e.g. WARRANTY_EXPIRED, GRACE_PERIOD, EXCLUDED_DAMAGE, SERIAL_MISMATCH, LATE_REPORTING, INCOMPLETE_EVIDENCE); which list a rule is in decides its severity, and a rule in no list is off. Every result — fired or passed — is stored with a message, so the explanation can show rules passed and failed. Files are validated against the vocabulary on load and on every admin save (atomic write, audited).

20. Python classification model design

Feature record (8 numeric, 10 binary, 2 categorical) → [ColumnTransformer](#) (scaler for the linear model, one-hot with fixed categories) → candidate classifier. Three candidates compared with stratified 5-fold CV and a validation split; the best on validation macro-F1 is wrapped in sigmoid calibration (5-fold) and saved as one pipeline artefact with its feature list and library version. Full evidence: [documentation/PYTHON_MODEL_EVIDENCE.md](#).

21. Google Teachable Machine model design

Standard image model trained on 2,100 Claim Summary Cards (700 per class; two variations of each training claim), exported as TensorFlow Lite. The card (v3) shows each claim fact against its policy limit as one of nine large colour-and-glyph tiles in a fixed 3 × 3 grid (coverage, reporting time, damage cause, receipt, supporting evidence, serial, dates, invoice, repairs), because Teachable Machine classifies with frozen ImageNet features and cannot read small text. Procedure and status: [documentation/GTM_EVIDENCE.md](#).

22. Dataset description

1,500 claims, exactly 500 per class, across three categories (Consumer Electronics 464, Home Appliances 514, Industrial Tools 522). Scenarios (from the ground-truth policy): covered defect 487, expired 223, missing documents 144, ambiguous cause 127, unauthorised repair 98, grace boundary 80, excluded damage 76, serial mismatch 60, contradiction 51, repeat repairs 35, possible duplicate 31, late reporting 23, plus 63 noise-flipped labels. This covers the SRS's normal, incomplete, contradictory, complex and borderline cases.

23. Dataset-generation method

[dataset_generator/generator_v2.py](#): attributes are sampled from realistic, overlapping distributions independently of the class (where in the warranty life the claim falls, reporting delay, damage cause by category, diagnosis, repairs, documents, serial match, duplicate invoice, date conflict). The label is then derived by an adjudication function that mirrors the policy files, and 4% of labels are flipped to model reviewer disagreement. Records are drawn until each class has 500, shuffled, and only then given claim IDs. A stratified 70/15/15 split follows, before any card is rendered. v1 had a [damage_type](#) column that was a renamed label and claim IDs ordered by class; both are fixed and tested.

24. Data pre-processing and feature engineering

Dates parsed with seven accepted formats; derived fields (product age, days to expiry, remaining warranty, reporting delay, missing-document count, repair count, unauthorised-repair / serial-mismatch / duplicate-invoice / date-conflict flags); missing documents → 0; unassessed diagnosis → 0.5; categorical values validated against the vocabulary and one-hot encoded with fixed categories; numeric scaling inside the pipeline for the linear candidate.

25. Python model-training procedure

[python notebooks/train_python_v2.py](#): leakage audit (fails above 90%) → 5-fold CV and validation metrics for each candidate → select on validation → permutation importance → calibrate → one-time test evaluation (accuracy, macro precision/recall/F1, ROC-AUC, log-loss, ECE, confusion matrix, per-class report, 30 sample predictions) → save pipeline and model card. Asserts accuracy $\geq 85\%$.

26. Google Teachable Machine training procedure

See [documentation/GTM_EVIDENCE.md](#) §“Training procedure”. After export the model is uploaded in Admin > Models, validated, versioned, and evaluated on the 450 held-out cards.

27. Model evaluation

Model	Accuracy	Macro precision	Macro recall	Macro F1	ROC-AUC
Python (HistGradientBoosting, calibrated), test	89.3%	89.5%	89.3%	89.2%	0.948
Teachable Machine (gtm-514702678b8c), test	93.8%	94.0%	93.8%	93.8%	—

28. Confusion matrix (Python model, test)

actual \ predicted	Invalid	Manual Review	Valid
Invalid Claim	61	8	6
Manual Review	2	69	4
Valid Claim	3	1	71

29. Accuracy, precision, recall and F1 by class (Python model, test)

Class	Precision	Recall	F1
Invalid Claim	92.4%	81.3%	86.5%
Manual Review	88.5%	92.0%	90.2%
Valid Claim	87.7%	94.7%	91.0%

30. Model prediction and confidence comparison

[reports/generate_comparison_report.py](#) produces [reports/model_comparison_report.csv/.md](#) for the test claims with every SRS column (IDs, actual class, both predictions and confidences, card filename, match, difference, status, rule result, missing documents, contradictions, duplicates, final decision, explanation of disagreements, summary). Consistency rules: minimum top confidence 0.60 (else Uncertain Result); different classes → Model Disagreement; $|\Delta| \leq 0.10$ Strong, ≤ 0.25 Acceptable, otherwise Weak Match; the latter three statuses and disagreement all route to manual review. On the test claims the two models agree on 92.9%. If the Teachable Machine export is missing, the report states that its columns are unavailable rather than estimating them.

Every stored verdict also carries its explanation: the three inputs that moved the Python model most (each replaced by its typical training value) and the Claim Summary Card tiles that changed the image model's answer most when blanked out. Administrators can replay all 225 test claims under different agreement thresholds in the what-if simulator before changing them; applying a change saves a new decision-policy version with its simulated effect in the audit trail.

Design check with a simulated agreeing image model (not a reported metric): the rules + decision table map 93.3% of test claims to their labelled class, and 10 of the 15 remaining differences are the deliberately noisy labels — i.e. the combination is sound once the second model is in place.

31. Testing strategy

225 automated pytest tests across unit (rules, detectors, OCR patterns, consistency, RBAC policy), integration (full HTTP journeys through the real app, DB and file store), boundary, negative, security, database, model and demonstration cases; surprise-modification scenarios are tests too. A generated route × role matrix calls every guarded route (68 route/method pairs) as each role and signed out and compares the answer with [config/rbac.json](#). Accuracy guards re-render the test cards with the current code and fail if the image model's accuracy could drop. A separate browser suite (`python -m pytest -m e2e tests/e2e`, 13 tests) runs the main journeys on a live seeded server with the real models, checks accessibility with axe, keyboard use, reduced motion, phone width and session revocation, and fails on any console error. Catalogue: [documentation/TEST_CASES.md](#); screenshots in [docs/screenshots/](#).

32. Security considerations

Deny-by-default RBAC with record scopes and separation of duties; out-of-scope records return 404; user state reloaded per request with session versioning (role change/disable signs out everywhere); idle timeout; generic login errors, five-attempt lockout, session rotation, rate-limited sign-in; public sign-up limited to customers; forced password change for invited users; CSRF tokens on all forms; strict CSP, clickjacking and MIME-sniffing protection; uploads validated by magic bytes, stored under random names outside the web root and served only through permission checks; CSV formula-injection guard; parameterised queries via the ORM; secrets from the environment; append-only audit including denied access. Design and v1 loophole list: [documentation/RBAC_DESIGN.md](#).

33. Privacy considerations

Only data needed to decide a claim is collected. Each role sees only its scope; documents are never publicly addressable; exports require an explicit permission and are audited. The repository contains only fictitious demo data and synthetic images. Deployments should use HTTPS (secure cookies are on by default), restrict database backups, and define a retention period for closed claims and uploaded files.

34. Limitations

- The Teachable Machine model must be retrained in the browser whenever the card design or policy limits change.
- Synthetic training data; rare scenarios (late reporting) are learned less well; real data needs retraining.
- Image OCR requires Tesseract on the server.
- Duplicate text similarity is lexical ($\geq 90\%$ sequence match), not semantic.
- SQLite and in-memory rate limits suit a single server; use PostgreSQL/Redis for multiple workers.

35. Future enhancements

Retrain on real claims with periodic recalibration and drift monitoring; feed the reviewer-decision export (Admin > Overview) back into training; image-based damage detection on customer photos; semantic duplicate detection; email/SMS notifications; multi-tenant manufacturer support; REST API for service-center systems; database migrations with Alembic.